

Table 1. Primer sequences used for amplification of TCF7L2 rs12255372

| Primer  | Sequence (5'→3')     | Product size |
|---------|----------------------|--------------|
| Forward | CTGGAAACTAAGGCGTGAGG | 346 bp       |
| Reverse | GGGTCGATGTTGTTGAGCTT | 346 bp       |

Table 2. Characteristics of Study Participants

| Variable                | T2DM (n=111)     | Non-DM (n=60)    | T1DM (n=10)      | p-value |
|-------------------------|------------------|------------------|------------------|---------|
| Age (years)             | 40.3 ± 6.73      | 31.5 ± 7.18      | 24.8 ± 6.05      | <0.001  |
| Sex (M/F)               | 23/88            | 4/56             | 2/8              | 0.001   |
| BMI (kg/m²)             | 23.2 ± 1.40      | 23.6 ± 1.69      | 22.7 ± 1.83      | 0.008   |
| Fasting glucose (mg/dL) | 159 (142–193)    | 87 (82–91)       | 199 (190–210)    | <0.001  |
| HbA1c (%)               | 8.83 ± 1.68      | 5.1 ± 0.17       | 8.4 ± 1.14       | <0.001  |
| Insulin                 | 9.8 ± 10.40      | 7.36 ± 1.38      | 2.0 ± 0.27       | <0.001  |
| C-peptide (ng/mL)       | 2.18 (1.59–2.53) | 2.33 (1.67–2.55) | 0.30 (0.20–0.30) | <0.001  |
| HOMA-IR                 | 3.5 (2.8–3.9)    | 1.5 (1.3–1.7)    | 1.0 (0.9–1.1)    | <0.001  |

Table 3. Distribution of rs12255372 Genotypes and Alleles

| Genotype/Allele | T2DM (n=111) | Non-DM (n=60) | p-value |
|-----------------|--------------|---------------|---------|
| GG              | 107 (96.4%)  | 58 (96.7%)    | 0.93    |
| GT              | 4 (3.6%)     | 2 (3.3%)      |         |
| G allele        | 218 (98.2%)  | 118 (98.3%)   | 0.92    |
| T allele        | 4 (1.8%)     | 2 (1.7%)      |         |

Table 4. Association Between rs12255372 and T2DM

| Genotype/Allele | OR (95% CI)      | p-value |
|-----------------|------------------|---------|
| GG              | Reference        | 0.93    |
| GT              | 1.08 (0.19–6.16) |         |

Table 5. Comparison of C-Peptide Levels according to genotype

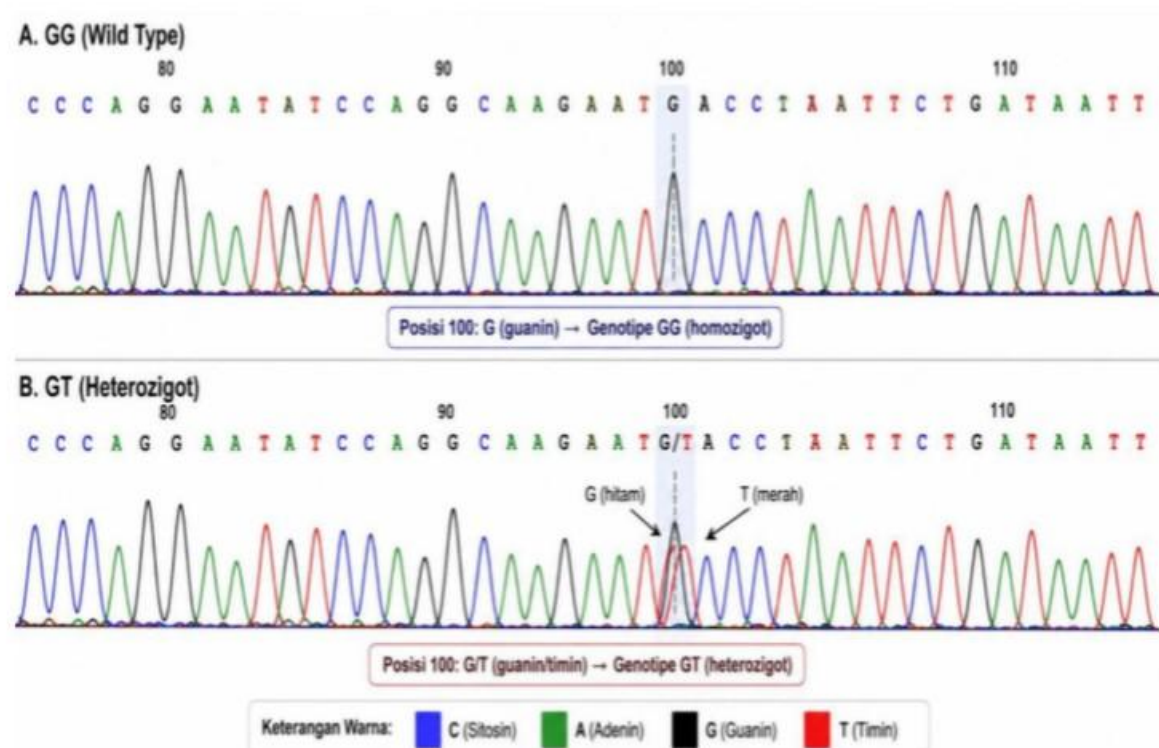
| Genotype   | Median C-peptide(IQR) ng/mL | p-value |
|------------|-----------------------------|---------|
| GG (n=175) | 2.1 (1.5–2.5)               | 0.157   |
| GT (n=6)   | 2.4 (2.1–2.6)               |         |

Table 6. Comparison of HOMA-IR according to genotype

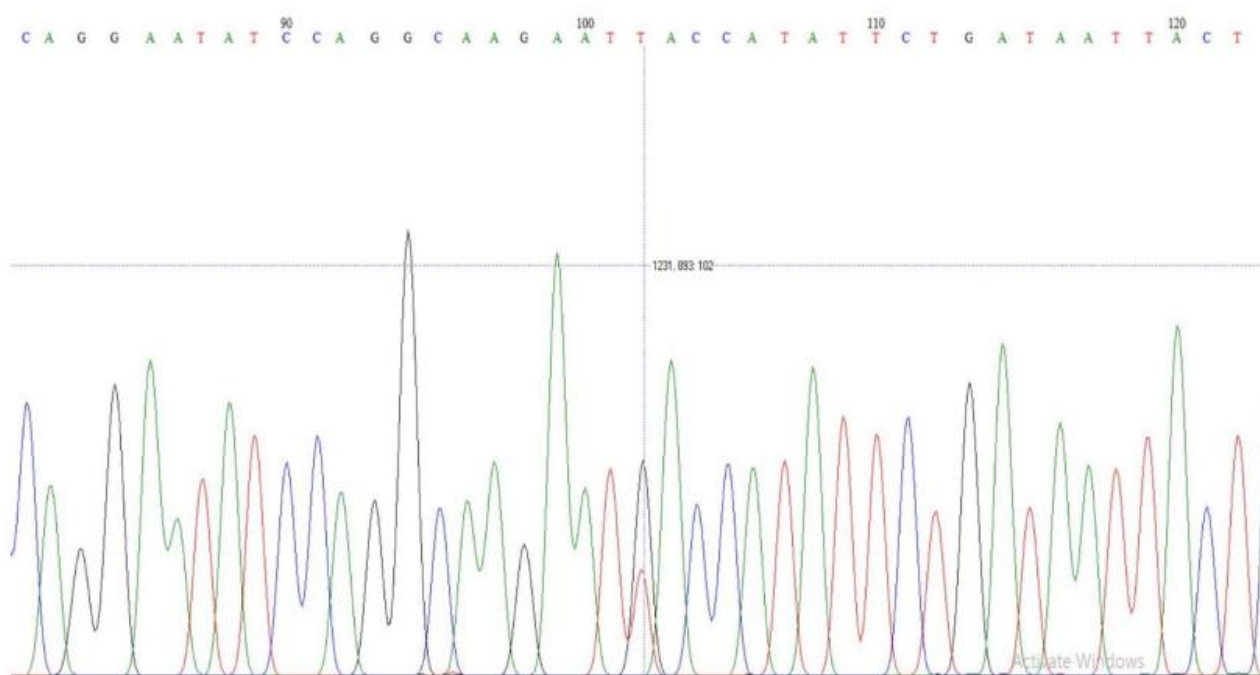
| Genotype   | Median HOMA-IR (IQR) | p-value |
|------------|----------------------|---------|
| GG (n=175) | 2.7 (1.6–3.6)        | 0.609   |
| GT (n=6)   | 3.0 (1.9–3.7)        |         |

Table 7. Multivariate Logistic Regression Analysis for T2DM

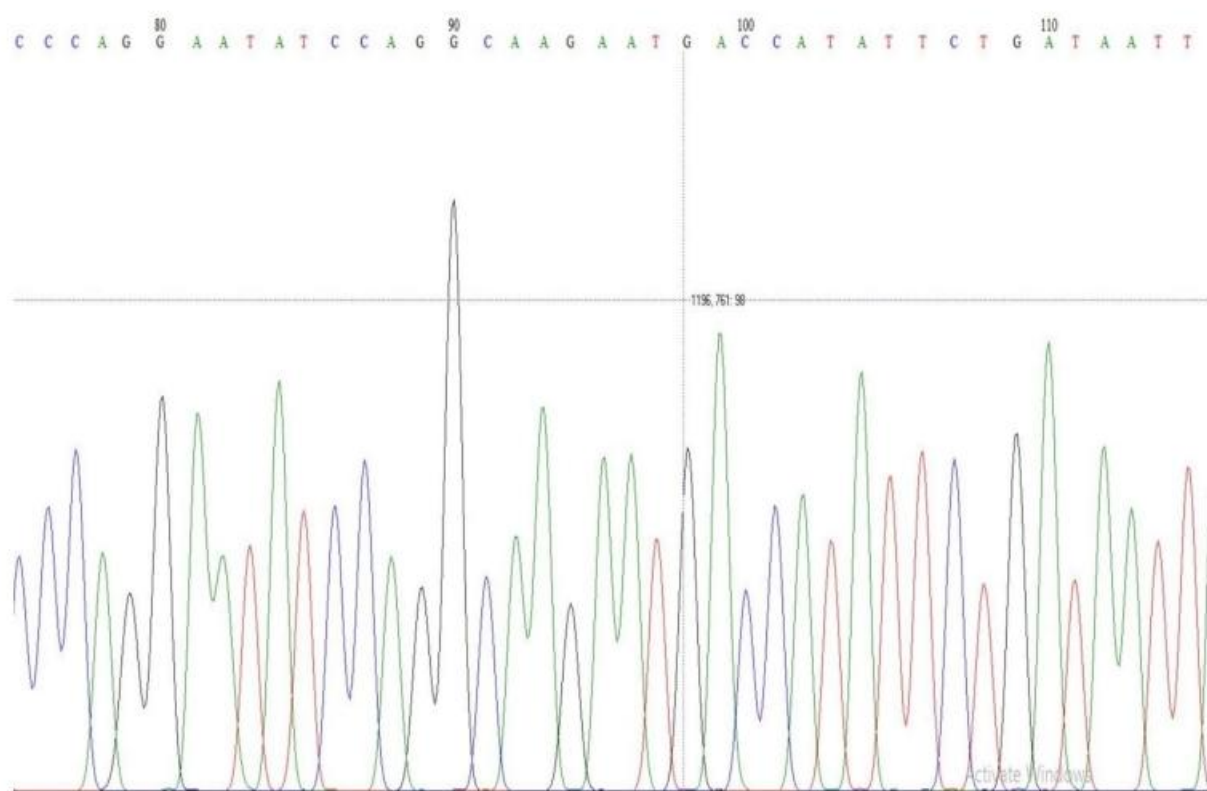
| Variable            | OR    | 95% CI      | p-value |
|---------------------|-------|-------------|---------|
| rs12255372 genotype | 0.646 | 0.070–5.969 | 0.700   |
| Age                 | 1.323 | 1.199–1.459 | <0.001  |
| BMI                 | 0.752 | 0.677–0.835 | <0.001  |
| Sex                 | 3.077 | 1.020–9.283 | 0.046   |



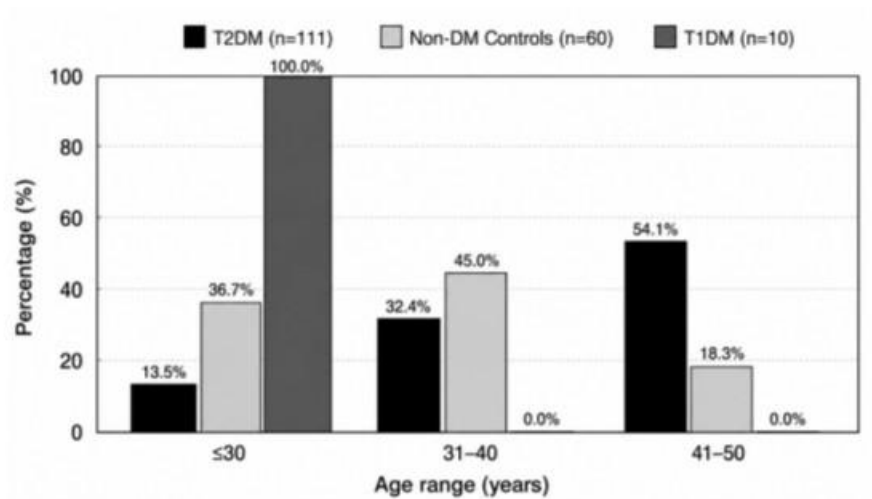
**Figure 4.** Summary of rs12255372 genotype patterns identified in this study. GG (wild-type homozygote) and GT (heterozygote).



**Figure 3.** Representative Sanger sequencing chromatogram of the rs12255372 GT genotype, showing overlapping guanine (G) and thymine (T) peaks at the polymorphic locus.



**Figure 2.** Representative Sanger sequencing chromatogram of the rs12255372 GG genotype, showing a single guanine (G) peak at the polymorphic locus.



**Figure 1.** Age-group distribution of participants with T2DM, non-diabetic controls, and T1DM.